STAT2402: Analysis of Observations

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Life lessons

The two most important days of your life are

the day you are born

and

the day your realise why.

Mark Twain

Part I: Introduction

Week 1

- Introduction to R
- RStudio
- RMarkdown
- Revision of Linear Statistical Models

“Analysis of Observations”?

- The unit should more correctly be titled ”Analysis of count data”.
- We will focus on the analysis of dichotomous (binary) and count response variables.
- Two types of discrete data:
    1. Dichotomous or binary data. Analysed as binomial tests.
    2. Count data. Analysed as contingency tables.

The simple techniques above do not allow for including the effect of other covariates.
A more general method is Generalised Linear Models (GLMs).

Unit Aims and Objectives

Specific learning objectives are:

1. demonstrate their knowledge of fundamental concepts in probability and statistics;
2. apply statistical models to real-world problems for discrete data, in particular count data;
3. use computer package(s) for fitting such models to data;
4. communicate the results of these analyses effectively to non-statisticians.

Mathematical knowledge

Advanced mathematics not assumed. However, comfort with some concepts are essential for understanding the modelling process. These are:

- Basic probability, such as conditional probability, independence etc.;
- The linear model in Gaussian regression, already covered in weeks 1 and 2;
- Exponents and logarithms;
- Basic calculus, in particular, an understanding of the process of finding maxima and minima of a function.

If you feel unfamiliar with any of the above then there are support materials available in *Statistical learning resources*.

Reading Material

- Applied Statistics:
  Roback, P. and Legler, J. (2021) *Beyond Multiple Linear Models and Multilevel Models in R*. CRC Press, Boca Raton, Fl.
- Theoretical Statistics:
  McCullagh, P., Nelder, J. (1989) Generalized linear models London : Chapman and Hall.

  Pawitan, Y. (2001). *In All Likelihood: Statistical Modelling and Inference Using Likelihood*, Oxford University Press, Oxford.
- Introduction to probability:
  Ross, S. (1988). *A First Course in Probability*, 3 edn, Macmillan Publishing Company, New York.

Recommended Reading

- The main reference book for the the course is:

  Faraway, J. J. (2016) *Extending the Linear Model with R: Generalized Linear, Mixed Effects and Nonparametric Regression Models*, 2nd Edition.
  You should get a copy if you can afford it. Note **the first edition is available for download online, free.**
- Also highly recommended reading is:

  Ramsey, F. and Schafer, D. (2013) *The Statistical Sleuth: A Course in Methods of Data Analysis*, 3rd Edition.

  Held in the High Demand section of the Barry J Marshall Library.
- Both books have `R` support. Ramsey and Schafer has on online repository of the `R` scripts and data sets for the book.

Contents: Weeks 3–8

- We will cover the following (broad references to chapters) chapters from the two books:
  - Proportions: Sleuth, Chapter 18.
  - Contingency tables: Faraway, Chapter 4; Sleuth, Chapter 19.
  - Comparison of proportions and odds: Sleuth, Chapter 18.
  - Logistic Regression: Faraway, Chapter 2; Sleuth, Chapter 20.
  - Logistic Regression for Binomial data: Faraway, Chapter 2; Sleuth, Chapters 20, 21.
- Additional material will be provided in lectures that introduces statistical concepts around sampling distributions and inference.

Weeks 9–11

- Log-Linear Regression for count data (Poisson): Faraway, Chapter 3; Sleuth, Chapter 22.
- Negative binomial regression.
- Zero-inflated Poisson regression.

Other

- The lecture notes will provide all of the information you will need on:
  - Sampling distributions and statistical inference
  - Likelihood and maximum likelihood estimation
  - Computational techniques for maximum likelihood estimation
  - Properties of maximum likelihood estimates
- The lecture notes should suffice but further material can be found in Pawitan (2001) and McCullagh and Nelder (1989).

Unit Outline

Read the Unit Outline carefully. In particular, assessments.

1. Weekly online quizzes, worth 2% each. These will be available on LMS each week, and are due by Sunday of the following week. The best ten quizzes will contribute to the final mark.
2. Two short tests, worth 10% each.
3. A mid-semester exam worth 15%.
4. Final examination, worth 45%.

R and RStudio

- Most common software for data analysis.
- Free and open source.
- Download and install the latest version on your laptop.
- Also download and install `Rstudio` on your laptop.
- We will demonstrate use of `R` in lectures.
- The Tuesday 2 pm session in Alexander LT will cover more details of `R`, in particular the `tidyverse` and `ggplot` packages.
- Later we will cover `RMarkdown` which will make preparing documents and reports much easier.
- *You will be using your own laptops in the lab classes.*

The Normal Distribution

Learning Outcomes

At the end of this chapter you should be able to:

1. understand the properties of a normal random variable;
2. compute probabilities for a normal distribution using the standard normal distribution tables and also R;
3. determine the distribution of sums of normal random variables;
4. model and solve problems using normal random variables;
5. be able to model and solve problems using a combination of continuous and discrete random variables as appropriate.

Introduction

A very important continuous distribution.

- Let the random variable X have a normal distribution with mean μ and variance σ^2. We write $X \sim N(\mu, \sigma^2)$. Note that the variance is given in this description.
- Let $Y = aX + b$. Then

$$\mathbb{E}(Y) = a\mu + b, \quad \mathrm{Var(Y) = a^2}\sigma^2,$$

and Y also has a normal distribution,
$Y \sim N(a\mu + b, a^2\sigma^2)$.

Standardisation

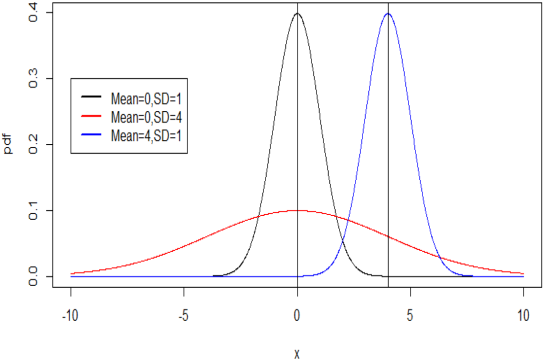
- In particular, if

$$Z = \frac{X-\mu}{\sigma} = \frac{1}{\sigma}X - \frac{\mu}{\sigma} \quad \left(a = \frac{1}{\sigma}, b = -\frac{\mu}{\sigma}\right)$$

  then $\mathbb{E}(Z) = 0$ and $\mathrm{Var(Z)} = 1$, so $Z \sim N(0,1)$. We call Z the standard normal distribution.
- If $Z \sim N(0,1)$ and $X = \mu + \sigma Z$, then $X \sim N(\mu, \sigma^2)$.

Probability density function — effect of changing the mean and variance

The mean is the central location, and variance or standard deviation is a measure of spread.



Normal probabilities

All probabilities will be computed using `R`.

Normal Distribution Problems

For $X \sim \mathrm{N}(\mu, \sigma^2)$, there are usually two types of problems.

- Given an interval for X, find the probability.
- Given a probability, find the value(s) of x (the **inverse** problem).

The examples below illustrate the ideas.

Example 1

Let $Z \sim \mathrm{N}(0, 1)$. Determine the following.

(i) $P(Z < 1.0)$

(ii) $P(Z < -1)$

Example 1 (ctd)

(iii) $P(Z > 1.0)$

(iv) $P(Z < 2.51)$

Example 1 (ctd)

(v) $P(Z < 1.96)$

(vi) The value of z such that $P(Z < z) = 0.9505$

Example 1 (ctd)

(vii) The value of z such that $P(Z < z) = 0.95$

(viii) $P(-1.5 < Z < 2.7)$

Example 2

Let $X \sim \mathrm{N}(5, 16)$. Determine the following.

(i) $P(X < 0)$

Example 2 (ctd)

(ii) $P(X > 10)$

(iii) $P(-5 \leq X \leq 7)$

Example 2 (ctd)

(iv) The value of c such that $P(X - \mu < c) = 0.95$

Example 3

Let $X \sim \mathrm{N}(\mu, 0.25)$, and suppose $P(X < 5.1) = 0.9772$. Find the value of μ.

Example 4

A machine fills bottles of soft drink to a mean volume of 210 mL with a standard deviation of 10 mL. The label on the bottle specifies a volume of 200 mL. A bottle is under-filled if it contains less than the labelled volume. Assume that the volumes of the bottles are normally distributed.

(a) What percentage of bottles are under-filled?

Example 4 (ctd)

(b) In order to reduce the percentage of under-filled bottles to 1% the company decides to adjust the standard deviation of the volumes filled by the machine. What should the standard deviation be reduced to?

SUM OF NORMAL RANDOM VARIABLES

Let $X \sim \mathrm{N}(\mu_{\mathrm{X}}, \sigma^2_{\mathrm{X}})$, $Y \sim \mathrm{N}(\mu_{\mathrm{Y}}, \sigma^2_{\mathrm{Y}})$, and put $W = X \pm Y$. Then $W \sim \mathrm{N}(\mu_{\mathrm{W}}, \sigma^2_{\mathrm{W}})$, where

$$\begin{aligned} \mu_W &= \mu_X \pm \mu_Y \\ \text{and} \quad \sigma^2_{\mathrm{W}} &= \mathrm{Var}(\mathrm{X} \pm \mathrm{Y}) \\ &= \mathrm{Var}(\mathrm{X}) + \mathrm{Var}(\mathrm{Y}) \pm 2(\mathrm{X}, \mathrm{Y}). \end{aligned}$$

If X and Y are independent, then $(X, Y) = 0$, so

$$\begin{aligned} \mu_W &= \mu_X \pm \mu_Y \\ \text{and} \quad \sigma^2_W &= \sigma^2_X + \sigma^2_Y. \end{aligned}$$

This result can be extended to a sum of several normal random variables.

Result

Let $X_1, X_2, \cdots, X_n$ be independent normal random variables with mean $\mathbb{E}(X_i) = \mu_i$ and $\mathrm{Var(X_i)} = \sigma_{\mathrm{i}}^2$, $i = 1, 2, \cdots, n$. Put

$$Y = \sum_{i=1}^{n} X_i.$$

Then $Y \sim N(\mu_Y, \sigma_Y^2)$, where

$$\mu_Y = \mathbb{E}\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} \mathbb{E}\left(X_i\right) = \sum_{i=1}^{n} \mu_i$$

and

$$\sigma_Y^2 = \mathrm{Var}\left(\sum_{\mathrm{i=1}}^{\mathrm{n}} \mathrm{X_i}\right) = \sum_{\mathrm{i=1}}^{\mathrm{n}} \mathrm{Var}\left(\mathrm{X_i}\right) = \sum_{\mathrm{i=1}}^{\mathrm{n}} \sigma_{\mathrm{i}}^2.$$

Example 5

Suppose the income of executives is normally distributed with the incomes of men and women independent of each other. The means and standard deviations (in \$10,000) respectively are 20, 2.5 for men and 15, 2.0 for women.

(a) What is the probability that a randomly chosen female executive earns more than a randomly chosen male executive?

Example 5 (ctd)

Example 5 (ctd)

(b) Find the probability that an executive couple has a combined income of more than \$400,000.

Example 6

A machine makes washers with hole diameters that are normally distributed, with mean 15.2 mm and variance 0.03 mm^2. Another machine makes bolts with diameters that are normally distributed, with mean 15.0 mm and variance 0.01 mm^2.

(a) What is the probability that a randomly selected bolt with fit through a randomly selected washer?

Example 6 (ctd)

(b) What should be the mean diameter of the washer holes if 99% of the bolts are to fit the washers?

Critical values of the standard normal distribution

We often need the value of the Z-distribution corresponding to given tail probabilities.

Example Find the value of z such that

1. $P(Z > z) = 0.01$

2. $P(Z < z = 0.05)$

3. $P(Z > z = 0.05)$

4. $P(Z > z = 0.01)$